



RESEARCH PROTOCOL & TECHNICAL SPECIFICATION

Skyhaven

A gamified behavioural assessment battery for measuring reactive and proactive aggression indicators in children and adolescents aged 8–16.

This document specifies what the game does, exactly what data each task collects, how every raw signal becomes a scored behavioural indicator, what the research evidence supports, what the earlier design got wrong, and how the corrected design must be validated before any interpretation is made about an individual child.

Version 1.0 · Draft for research review

Not a diagnostic instrument

DOCUMENT OWNER
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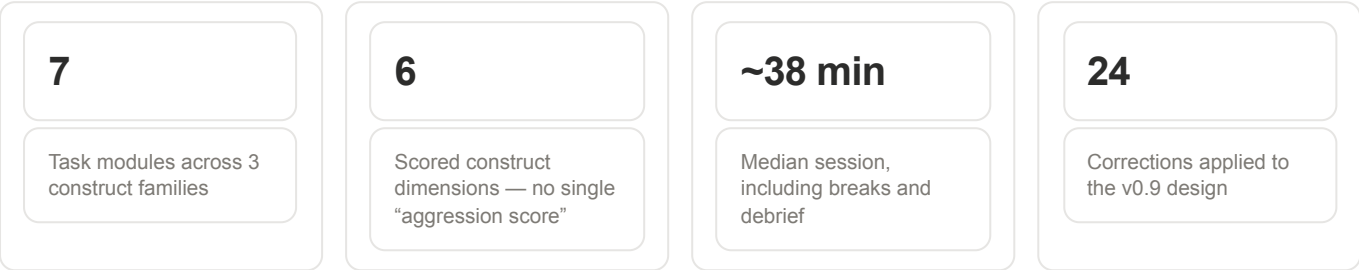
DATE
31 August 2026

SUPERSEDES
Task table v0.9

STATUS
Pre-validation

1 Executive summary

Skyhaven is a supervised, session-based assessment battery presented to the child as a seven-stage game. Every stage is an adaptation of an established experimental paradigm from the aggression and executive-function literature. The child experiences a game; the system records a time-stamped behavioural event stream. Nothing is inferred from what the child *says* about themselves alone — every construct is anchored to observable behaviour (latency, error, choice, persistence) plus a small number of pictorial affect ratings.



1.1 What the battery is, and what it is not

SKYHAVEN IS	SKYHAVEN IS NOT
A research instrument that produces a profile of behavioural indicators associated in the literature with reactive and proactive aggression.	A diagnostic test. It cannot identify conduct disorder, ODD, ADHD, or “an aggressive child”.
An objective complement to teacher, parent and self-report measures, reducing reliance on social desirability.	A replacement for those measures. It must be validated <i>against</i> them.
A controlled, pre-registered, ethically bounded frustration and provocation environment.	A deception experiment, a stress test, or a screening tool for schools to rank students.
Fully instrumented: every score traces back to raw, auditable events.	A black box. No machine-learning classifier is used in v1.0, by design.

GOVERNING CONSTRAINT

Until the validation study in §8 is completed and reported, Skyhaven outputs are **group-level research data only**. No individual-level feedback, label, risk flag or referral may be generated from a session. This constraint is enforced in software: individual report generation is disabled behind a feature flag until norms exist.

1.2 Document map

2	Correction log — what was wrong in v0.9 and how it was fixed	Design integrity
3	Theoretical framework and construct map	Why each task exists
4	Session architecture and flow control	Order, washout, safety gates
5	Module specifications M1–M7	Mechanics, events, metrics, evidence
6	Data architecture — what the backend actually stores	Event schema and pipeline
7	Scoring pipeline and researcher-facing output	Raw → feature → construct
8	Psychometric validation plan	Reliability, validity, sample
9	Ethics, consent, safeguarding, data protection	Approval requirements

10 Limitations, confounds and known threats to validity

Honest boundaries

Appendix A–C Formula reference, data-quality rules, references

Implementation detail

2 Correction log: what v0.9 got wrong

The original task table was directionally sound — the paradigms chosen are the right ones and the cited literature genuinely supports them. However, as written it would not have produced interpretable data. The issues below are ordered by severity. Every one is corrected in this specification.

2.1 Critical — would invalidate the measurement

ISSUE IN V0.9	WHY IT BREAKS THE MEASUREMENT	CORRECTION IN V1.0
C1. Hostile attribution measured with intentional vs. accidental framing	This is the single most important error. Hostile attribution bias is, by definition, the tendency to read hostile intent into ambiguous provocation. If you tell the child the act was intentional, an angry response is <i>accurate social cognition</i> , not bias. v0.9 would have measured comprehension, not bias.	Three-condition vignette design: ambiguous (the scored condition), plus clearly hostile and clearly benign versions used only as comprehension and validity checks. HAB is scored from the ambiguous condition alone (M6).
C2. No baseline / neutral block	Without a pre-frustration baseline you cannot separate “this child is slow and error-prone in general” from “this child deteriorates when frustrated”. The frustration effect is a change score and v0.9 had nothing to change from.	Mandatory neutral baseline block before any frustration induction, and a neutral washout block after it. All frustration and provocation metrics are reported as deltas from baseline.
C3. Stop-signal delay not adaptive	A stop signal that “occasionally appears” at a fixed delay produces an uninterpretable inhibition estimate. SSRT can only be computed when the delay is tracked so that inhibition succeeds on roughly half of stop trials.	1-up/1-down staircase on stop-signal delay (± 50 ms, converging on $p(\text{inhibit}) \approx 0.50$), SSRT estimated by the integration method with replacement of go omissions (M1).
C4. Go/No-Go ratio unspecified	Inhibition is only taxed if the “go” response becomes prepotent. At a 50:50 ratio there is nothing to inhibit, and commission errors stop being an impulsivity index.	Fixed 75:25 go:no-go ratio, minimum 120 go and 40 no-go trials per emotional condition for stable estimates.
C5. Too few trials for reliable estimates	Reaction-time and error metrics from a handful of trials have near-zero split-half reliability. Individual differences cannot be read from noisy estimates, however good the dashboard looks.	Per-cell trial minimums specified for every module (§5), with split-half reliability computed per session and sessions below $\rho = 0.60$ flagged as unusable.
C6. Device and input latency ignored	Touch latency, refresh rate and browser scheduling differ by 40–120 ms across devices. Comparing raw RTs across children on different tablets measures the hardware, not the child.	Calibration routine at session start; device class, refresh rate and measured input lag stored per session; RT analyses use within-subject contrasts and include device class as a covariate (§6.4).
C7. Impulsivity treated as specific to aggression	Poor inhibition is common to ADHD, anxiety, sleep deprivation, low motivation and age. Reading it as an aggression indicator on its own would generate large numbers of false positives.	Co-collection of a brief screening measure (SDQ) so hyperactivity/inattention can be modelled as a covariate. Inhibition is reported as its own dimension, never folded into an aggression score.

2.2 Serious — would bias or contaminate results

ISSUE IN V0.9	PROBLEM	CORRECTION
C8. Explicit aggressive options (“Insult them”, “Get even”)	Labelled aggressive choices invite social-desirability suppression: children learn within two items which button is the “bad” one and stop selecting it, compressing variance to near zero.	Options phrased as concrete in-world actions rather than moral labels, presented in randomised order, with decision latency logged. A social-desirability index (M6) is computed and used as a covariate.
C9. Carryover between frustration and later blocks	Anger induced in the puzzle module persists for minutes and inflates later provocation scores, so modules are not independent measurements.	Mandatory 90-second neutral washout plus an affect-return check between induction modules; the next module does not start until self-reported arousal returns within 1 point of baseline or 3 minutes elapse.
C10. Fixed module order	A fixed order confounds construct with fatigue and practice.	Latin-square counterbalancing of module order within the constraint that baseline is always first and the retaliation module always last.
C11. “Impossible puzzle” framed as deception	Unsolvable trials presented as solvable is deception of a minor. It is defensible in adult research with debrief, but with children it raises approval and welfare problems — and v0.9 gave no exit condition.	Failure is scheduled, not deceptive: a fixed rigged-trial schedule with a hard 90-second cap, an always-visible exit control, no failure streak longer than two trials, and a scripted debrief explaining that some rounds were built to be unwinnable and that this says nothing about the child’s ability.
C12. Numeric Likert scales for young children	8–10 year olds use numeric scales inconsistently; the resulting affect data is noisy and age-confounded.	Pictorial 5-point Self-Assessment Manikin style scales for anger and arousal, identical at every measurement point.
C13. No attention or effort checks	A disengaged child who taps randomly produces data indistinguishable from an impulsive child.	Embedded attention probes (≥ 1 per module), anticipatory-response detection, and an engagement index; sessions failing $>20\%$ of probes are excluded (Appendix B).
C14. Emotional context absent from the Go/No-Go	The study v0.9 cites for this task used an emotional Go/No-Go with angry faces; the impairment found in reactively aggressive adolescents was specific to angry-face conditions. A neutral shapes task does not test that finding.	M1 uses validated child facial-affect stimuli in three blocks (angry / happy / neutral). The scored contrast is inhibition on angry versus neutral blocks.
C15. Persistence not operationalised	“Quitting behaviour” was named but never defined, so it could not be computed.	Three explicit persistence metrics: time-to-first-quit, attempts before quit, and area under the effort curve (Appendix A).
C16. No provision for reading ability	Text-heavy vignettes measure reading fluency as much as social cognition, penalising younger and weaker readers.	All vignettes are audio-narrated with illustration, text below Flesch–Kincaid grade 3, and comprehension checks per vignette.

2.3 Corrections applied silently across the specification

C17 randomised stimulus order and pseudo-random inter-trial intervals to prevent rhythmic responding · C18 practice blocks with feedback before every scored block · C19 explicit data-quality exclusion rules rather than post-hoc judgement · C20 versioned scoring code so a score can always be reproduced from stored events · C21 pseudonymised participant identifiers separated from the identity table · C22 the optional retaliation module gated behind separate ethics approval and separate

parental consent · C23 no single composite “aggression score” is produced · C24 pre-registration of hypotheses and analysis plan before data collection.

NET EFFECT OF THE CORRECTIONS

v0.9 would have generated a large, attractive, and largely uninterpretable dataset. v1.0 generates a smaller number of defensible quantities, each with a stated computation, a stated confound, and a stated evidence base. A reviewer can disagree with an interpretation but cannot be confused about what was measured.

3 Theoretical framework and construct map

Skyhaven is built on three literatures, each contributing a distinct construct family. Keeping them separate is deliberate: they are empirically distinguishable and conflating them is how gamified assessments become pseudoscience.

3.1 The three construct families

FAMILY	THEORY	CORE CLAIM	WHAT SKYHAVEN MEASURES FROM IT
A. Inhibitory control	Response-inhibition / horse-race model of stopping	Aggressive acts in the reactive subtype are frequently failures to suppress a prepotent response under emotional load.	Commission errors, SSRT, RT variability, and crucially the angry-minus-neutral inhibition cost.
B. Frustration and negative affect	Berkowitz’s cognitive-neoassociationist reformulation of frustration–aggression	Goal blockage produces negative affect, which primes aggressive inclinations — but the outcome depends on interpretation and context, so aggression is one possible response among several.	Frustration tolerance, persistence, disengagement latency, and the range of responses selected after blockage.
C. Social information processing	Crick & Dodge SIP model; hostile attribution bias	Children differ systematically at each processing step: encoding cues, interpreting intent, generating responses, evaluating outcomes, and enacting.	Hostile intent attribution under ambiguity, response repertoire, evaluated outcome of aggression, retaliation severity.

3.2 Reactive versus proactive: why both are separated

The two subtypes have different correlates and are targeted by different interventions. A design that produces one blended score is less useful than one that keeps them apart.

	REACTIVE (HOT, DEFENSIVE)	PROACTIVE (COLD, INSTRUMENTAL)
Trigger	Perceived threat, provocation, frustration	Anticipated reward or goal
Affect	High anger and arousal	Low arousal, planned
SIP locus	Encoding and interpretation (hostile attribution)	Response evaluation (expects aggression to pay off)
Skyhaven signal	M1 angry-block inhibition cost, M3/M5 frustration deltas, M6 HAB index	M7 repeat-choice after neutral consequence, M6 expected-outcome ratings, low-arousal severity choices

3.3 Construct-to-module matrix

SCORED DIMENSION	PRIMARY MODULE	CONVERGING MODULE	PRIMARY METRIC
D1 Inhibitory control	M1	M2	SSRT, commission error rate

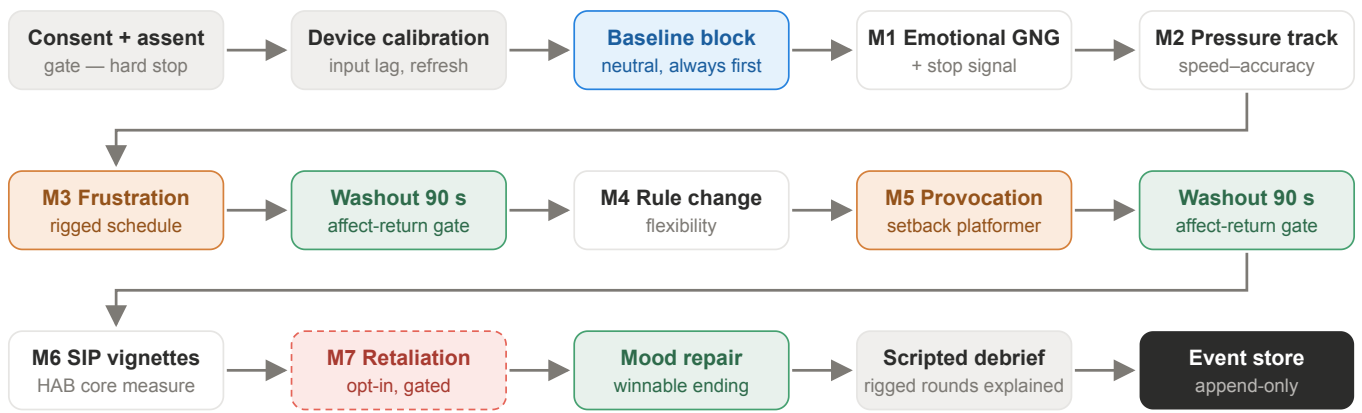
D2 Emotional inhibition cost	M1	M5	Angry-block minus neutral-block commission rate
D3 Frustration tolerance	M3	M4, M5	Effort AUC, time-to-quit delta
D4 Hostile attribution bias	M6	M5	Hostile-intent proportion, ambiguous condition
D5 Retaliatory tendency	M6	M5, M7	Weighted severity index
D6 Outcome expectancy / persistence of aggression	M7	M6	Repeat rate after neutral consequence

DESIGN RULE

No dimension is scored from a single module without at least one converging measure. Where the primary and converging measures disagree by more than 1.5 SD, the dimension is reported as *inconsistent* rather than averaged. Averaging away a disagreement destroys the most informative thing in the data.

4 Session architecture and flow control

A session is a fixed pipeline with safety gates. The child sees a continuous adventure; the system sees eleven discrete phases with defined entry and exit conditions.



4.1 Phase gates

GATE	CONDITION TO PROCEED	IF THE CONDITION FAILS
Consent + assent	Parental/guardian consent on file and child assent recorded in-session	Session cannot start; nothing is logged beyond the refusal event
Calibration	Measured input lag < 150 ms, stable frame timing over 10 s	Session marked <code>rt_unreliable</code> ; choice-based modules still valid, RT modules flagged
Practice mastery	≥80% correct on practice trials, up to 3 practice rounds	Module skipped and marked <code>not_comprehended</code> rather than scored as poor performance
Affect return (washout)	Anger rating within 1 point of baseline, or 3-minute cap reached	Cap reached → next module tagged <code>carryover_risk</code> and delta-adjusted
Distress stop-rule	Anger/upset rating ≤ 4 of 5, no verbal distress flag from the supervising adult	Immediate stop , mood repair, debrief, welfare escalation per §9.4

ON THE “LEVEL DEVIL” REFERENCE IN V0.9

Trap-based troll games are entertaining precisely because they are unfair and unpredictable in unbounded ways. For measurement, unbounded unfairness is worthless: if the difficulty a child experiences is not identical across children, differences in reaction cannot be attributed to the children. M5 therefore uses a **fixed, pre-scripted setback schedule** — every child hits the same traps, at the same points, in the same order.

5 Module specifications

Each module below states: what the child experiences, the design parameters that make it measurable, the raw events written to the backend, the derived metrics with their formulas, how the result is read, and the evidence base. Metric identifiers are used verbatim in the scoring pipeline (§7) and Appendix A.

M1

Emotional Go/No-Go with Stop-Signal — “Signal Tower”

D1, D2 · ~9 min

WHAT THE CHILD DOES

Faces appear one at a time on the tower’s screen. Tap for the “go” face, hold back for the “no-go” face. On a minority of go trials an alarm sound fires shortly after the face — that means abort the tap already in motion. Runs in three themed blocks (angry, happy, neutral faces).

WHY IT MUST BE EMOTIONAL

The adolescent evidence base for reactive aggression found impaired inhibition specifically when angry expressions were being processed — not on neutral stimuli. A shapes-only task cannot detect the effect the design is built to detect.

DESIGN PARAMETER	VALUE AND RATIONALE
Go : No-Go ratio	75 : 25 — builds a prepotent response so withholding is genuinely effortful
Trials per block	160 (120 go, 40 no-go) × 3 emotion blocks; block order counterbalanced
Stop-signal trials	25% of go trials; delay adapts ±50 ms by 1-up/1-down staircase toward p(inhibit) = 0.50
Stimulus timing	Face 500 ms, response window 1000 ms, jittered inter-trial interval 800–1400 ms
Stimuli	Validated child/adolescent facial-affect set, balanced for sex and skin tone, licensed for research

RAW EVENTS WRITTEN

```
trial_start(trial_id, block, emotion, stim_id, is_nogo, is_stop, ssd_ms, t_onset)
response(trial_id, t_response, pointer_xy, hold_duration_ms)
trial_end(trial_id, outcome ∈ {hit, miss, correct_rejection, commission, stop_success, stop_failure})
```

DERIVED METRICS

METRIC ID	COMPUTATION	READS AS
m1_commission	commission errors ÷ no-go trials, per emotion block	Motor impulsivity / failure to withhold
m1_dprime	z(hit rate) – z(false-alarm rate), log-linear corrected	Discriminability, separates carelessness from inability
m1_ssrt	Integration method: nth go RT – mean SSD, where n = p(respond signal) × go-RT count	Speed of the internal stopping process, in ms
m1_rtcv	SD(go RT) ÷ mean(go RT)	Attentional consistency; high values indicate lapses
m1_pes	mean RT on trial n+1 after error – after correct	Error monitoring; near-zero suggests errors are not registering
m1_emo_cost	m1_commission(angry) – m1_commission(neutral)	Primary D2 index — inhibition breakdown under anger cues

INTERPRETATION GUARD

High m1_commission with high m1_rtcv and flat m1_pes is a **generalised attention** pattern, not an aggression signal. Only an elevated m1_emo_cost with intact neutral-block performance is specific to the reactive-aggression hypothesis.

M2 Pressure Tracking — “Sky Runner”

D1 converging · ~5 min

WHAT THE CHILD DOES

Intercept a moving target that speeds up in fixed increments across five levels, with scripted visual distractors on levels 3–5. Explicit instruction: accuracy matters more than speed.

MEASUREMENT PURPOSE

Not an aggression measure. It quantifies how a child trades speed against accuracy as load rises, which is the covariate needed to interpret M1 and to distinguish “fast and careless” from “fast and capable”.

DESIGN PARAMETER	VALUE
Levels	5, fixed speed multipliers 1.0 / 1.25 / 1.5 / 1.8 / 2.2 — identical for every child, never adaptive
Trials per level	20 (100 total)
Distractors	Scripted, identical timing for all participants

DERIVED METRICS

METRIC ID	COMPUTATION	READS AS
m2_lisas	mean RT + $(SD_{RT} \div SD_{PE}) \times$ proportion error, per level	Integrated speed–accuracy performance
m2_decay	OLS slope of m2_lisas across the five levels	Performance degradation under increasing pressure
m2_strategy	Δ accuracy $\div$ Δ RT between level 1 and level 5	Whether the child sacrifices accuracy for speed when pushed

M3 Scheduled Frustration — “The Locked Gate”

D3 · ~7 min

WHAT THE CHILD DOES

Eight timed tile puzzles open the gate. Trials 3, 5 and 6 are rigged: the required piece is unreachable within the time limit. A “Skip this gate” control is visible at all times from trial 1. Anger and arousal are rated on a pictorial scale after trials 2, 6 and 8.

ETHICAL CONSTRUCTION

No streak longer than two rigged trials. Hard 90-second cap per trial. Skipping costs nothing in the game economy. The debrief states plainly that some gates could not be opened by anyone.

METRIC ID	COMPUTATION	READS AS
m3_ttg	Median seconds from trial start to skip/quit press on rigged trials	Disengagement latency — the core persistence measure
m3_effort_auc	Area under the cumulative-interaction curve over the rigged trials, normalised by cap	Sustained effort against blockage
m3_retry	Retry presses $\div$ available retry opportunities	Behavioural persistence preference
m3_anger_delta	Anger rating after trial 6 – baseline rating	Affective reactivity to goal blockage
m3_input_force	Tap rate and inter-tap interval variance during rigged vs. fair trials	Agitated input; weak signal, reported but never scored alone

DIRECTION IS NOT OBVIOUS

Both very fast quitting and rigid over-persistence can indicate poor frustration regulation. `m3_ttq` must be interpreted against the group distribution, not as “higher is better”. Berkowitz’s reformulation is explicit that frustration produces negative affect with several possible behavioural outcomes — aggression is not the default.

M4

Rule-Change Adaptation — “Shifting Bridges”

D3 converging · ~4 min

Rules governing which bridge tile is safe change silently every 12 trials; the child must detect the change and switch strategy. This is a set-shifting task, and it disambiguates M3: a child who quits the locked gate quickly but adapts smoothly here is *efficient*, whereas one who quits quickly and also perseverates here shows a genuine flexibility deficit.

METRIC ID	COMPUTATION	READS AS
m4_switch_cost	Mean RT on the 3 trials after a rule change – the 3 before	Cognitive flexibility cost
m4_persev	Consecutive old-rule responses after the change point	Perseveration / rigidity
m4_recovery	Trials required to reach 3 consecutive correct under the new rule	Adaptation speed

WHAT THE CHILD DOES

A side-scrolling course in which an on-screen rival character causes six scripted setbacks (blocks a path, takes a collected item, triggers a fall). After each setback the child chooses one of five in-world actions before continuing.

WHY AN AGENT, NOT AN OBSTACLE

Reactive aggression is defined by response to **perceived threat or provocation from an agent**. An impersonal trap tests frustration only. A visible rival makes intent attribution possible, which is what the construct requires.

RESPONSE OPTIONS AND SEVERITY WEIGHTS

IN-WORLD ACTION SHOWN TO THE CHILD	CODED CATEGORY	WEIGHT W	NOTE
“Find another way around”	Adaptive / avoidant	0	Task-focused
“Ask the guide for help”	Help-seeking	0	Prosocial
“Tell the rival to stop”	Assertive verbal	1	Non-aggressive assertion
“Block their path back”	Instrumental retaliation	2	Property/goal directed
“Knock their lantern down”	Direct retaliation	3	Costly, no task benefit

Options are worded as concrete world actions, never as moral labels, and their on-screen order is randomised per trial to prevent position bias. Decision latency is logged: fast severe choices and slow severe choices are theoretically different (impulsive versus deliberate) and are analysed separately.

METRIC ID	COMPUTATION	READS AS
m5_severity	$\sum w \div (3 \times \text{number of provocation events})$, range 0–1	Retaliatory tendency under provocation
m5_escalation	OLS slope of w across the six ordered setbacks	Whether responses harden with repeated provocation
m5_latency	Median decision time for $w \geq 2$ choices	Impulsive vs. deliberate retaliation
m5_repertoire	Count of distinct categories used across the six events	Response flexibility; a narrow repertoire is itself a SIP finding

M6

Social Information Processing Vignettes — “Village Stories”

D4, D5 · ~9 min · highest-value module

Twelve illustrated, audio-narrated scenarios covering exclusion, teasing, unfair blame, lost belongings, competition and the “stolen credit” case from v0.9. Each scenario exists in three intent versions; each child sees each scenario once, with version assignment counterbalanced across the sample.

INTENT CONDITION	SHARE	FUNCTION
Ambiguous	6 of 12	The scored condition. Cause of the negative outcome is genuinely undetermined by the narration and illustration.
Clearly hostile	3 of 12	Validity check — a child who does <i>not</i> attribute hostility here has not understood the task, and their ambiguous-condition data is discarded.
Clearly benign	3 of 12	Validity check — hostile attribution here indicates indiscriminate responding or extreme bias; flagged separately.

QUESTIONS ASKED PER VIGNETTE (FIXED ORDER, SIP-STEP ALIGNED)

SIP STEP	QUESTION	RESPONSE FORMAT
Encoding	“What happened in the story?”	3-option comprehension check (gate)
Intent interpretation	“Why did they do it?”	Hostile / benign / accidental / unsure
Affect	“How angry would you feel?”	Pictorial 5-point scale
Response generation	“What would you do?”	Open-ended (voice or text), coded post-hoc
Response selection	Choice among 5 weighted actions	Forced choice, randomised order
Outcome evaluation	“If you did that, what would happen next?”	Better / worse / no change + confidence

METRIC ID	COMPUTATION	READS AS
m6_hab	Hostile intent attributions ÷ ambiguous vignettes (range 0–1)	Primary D4 index. The best-evidenced quantity in the battery.
m6_anger_amb	Mean anger rating, ambiguous condition	Anger sensitivity under uncertainty
m6_severity	Weighted severity index across all scored vignettes	Aggressive response selection
m6_expectancy	Proportion of aggressive selections rated as producing a better outcome	Proactive marker — aggression evaluated as effective
m6_repertoire	Distinct solution types generated in the open-ended step	Breadth of the response repertoire
m6_sdr	Divergence between open-ended generation and forced-choice selection	Social-desirability index, used as a covariate

EVIDENCE STRENGTH

This module carries the strongest empirical backing in the battery. A large meta-analysis of hostile intent attribution and aggression in children and adolescents (111 studies, 29,272 participants) reported a robust positive association, and a systematic review of 27 studies found the association held particularly for the *reactive* subtype. The ambiguity requirement is not a stylistic preference — it is the operational definition used across that literature.

After a final scripted unfair loss, the child selects a response, sees a **neutral, non-reinforcing** consequence (the story simply continues; the rival is neither punished nor rewarded), then answers whether they would make the same choice again and whether it was fair. The neutrality is essential: a consequence that rewards retaliation teaches aggression, and a consequence that punishes it measures compliance rather than preference.

METRIC ID	COMPUTATION	READS AS
m7_repeat	Proportion of aggressive choices the child would repeat after a neutral outcome	Stability of retaliatory preference vs. one-off impulse
m7_selfeval	Self-rated fairness of own response, aggressive choices only	Moral evaluation / self-monitoring
m7_shift	Change in severity weight between first and repeated choice	Regulation after reflection

GATING CONDITIONS — ALL MUST HOLD

Separate line item in the parental consent form · child re-assent immediately before the module · independent ethics-committee sign-off · automatic skip for any child flagged by the M3/M5 distress rule · mood-repair sequence always follows · module omitted entirely for participants under 10 in the first study wave.

5.1 Coverage check against the v0.9 table

V0.9 ELEMENT	STATUS	WHERE IT LIVES NOW
1. Go/No-Go + Stop-Signal	Kept, fixed	M1 — emotional blocks, adaptive SSD, trial minimums
1B. Moving target reaction	Kept, re-scoped	M2 — reframed as a covariate measure, fixed difficulty ramp
2. Frustration puzzle	Kept, fixed	M3 — scheduled failure, exit control, pictorial affect scale
2B. Changing-rules puzzle	Kept	M4 — now explicitly a set-shifting disambiguator
3. Level-Devil-style challenge	Rebuilt	M5 — scripted identical setbacks, personified rival
3B. Provocation choice events	Kept, re-worded	M5 — neutral wording, randomised order, latency logged
4. Stolen credit case	Kept	M6 — one of twelve vignettes, three intent versions
4B. Multiple conflict stories	Kept, expanded	M6 — twelve scenarios for cross-situational consistency
4C. Intentional vs. accidental	Corrected	M6 — ambiguous condition scored; the other two are validity checks
5. Revenge decision task	Gated	M7 — opt-in, separately consented, neutral consequences
5B. Consequence awareness	Kept	M7 — repeat-choice and self-evaluation
—	Added	Baseline block, washouts, calibration, practice, attention probes, debrief

6 Data architecture: what the backend actually stores

The system is **event-sourced**. The client emits an append-only stream of immutable events; every metric in this document is a pure function of that stream. Nothing is computed on the device and nothing derived is ever the source of truth. This is what allows a reviewer to recompute any published number from the raw record.

6.1 Storage layers

LAYER	CONTENTS	MUTABILITY	PURPOSE
L0 Identity vault	Name, school, DOB, guardian contact, consent documents	Restricted, separate database	Never joined to behavioural data outside a controlled linkage process
L1 Event store	Every stimulus, response, choice, rating, pause, focus-loss	Append-only, immutable	Ground truth; the only thing that must never be lost
L2 Trial table	One row per trial, reconstructed from L1	Rebuildable	Analysis-ready flat structure
L3 Feature table	The ~30 metric IDs from §5, per participant per module	Rebuildable, versioned	Input to scoring
L4 Score table	Six construct dimensions, standardised	Rebuildable, versioned	Researcher output

6.2 Canonical event record

```
{
  "event_id":      "01J9F2K7X3QW8A",           // ULID, client-generated, idempotency key
  "session_id":    "ses_7f3c91",
  "participant_id": "pid_4a91e0",              // pseudonym; no identity in this store
  "module":        "M1",
  "block":         "angry",
  "trial_index":   37,
  "event_type":    "response",
  "t_client_mono": 184392.418,                 // performance.now(), monotonic, ms
  "t_client_wall": "2026-08-31T16:44:02.418Z",
  "t_server_recv": "2026-08-31T16:44:02.606Z", // clock-skew correction anchor
  "payload": {
    "stim_id": "face_ang_014", "is_nogo": true, "is_stop": false,
    "rt_ms": 291, "outcome": "commission", "pointer": [412, 638]
  },
  "quality": {
    "frame_jitter_ms": 3.1, "input_lag_ms": 62,
    "window_focused": true, "attention_probe_ok": true
  },
  "schema_version": "1.0.0"
}
```

6.3 Event taxonomy

EVENT TYPE	EMITTED WHEN	FEEDS
session_start / end	Session boundaries, with device and calibration data	Quality gating
consent_recorded	Guardian consent and child assent captured	Compliance audit

calibration_result	Input-lag and frame-timing measurement completes	RT correction, device covariate
trial_start / response / trial_end	Every experimental trial	All RT and error metrics
choice_made	Any forced-choice selection, with option order as displayed	Severity, escalation, latency
rating_given	Pictorial affect scales	Anger deltas, washout gate
open_response	Voice/text response generation	Repertoire coding
quit_pressed / retry_pressed	Persistence-relevant controls	Frustration metrics
focus_lost / focus_gained	App backgrounded or window blurred	Trial invalidation
distress_flag	Stop-rule triggered by rating or supervising adult	Safeguarding, exclusion
debrief_completed	Debrief script finished	Compliance audit

6.4 Timing integrity

THREAT	MITIGATION IMPLEMENTED
Device input latency differences	Per-session calibration; measured lag stored and used as a covariate; RTs never compared across device classes without adjustment
Wall-clock drift and skew	All within-trial timing from a monotonic clock; wall clock used only for ordering and anchored to server receipt time
Dropped frames / GC pauses	Frame jitter recorded per trial; trials with jitter > 20 ms excluded from RT metrics
Network loss mid-session	Offline-first buffering with ULID idempotency keys; replay on reconnect cannot duplicate events
App backgrounding	Focus events invalidate the affected trial and the following one

7 Scoring pipeline and researcher-facing output



7.1 Standardisation

Raw metrics are meaningless in isolation: 280 ms SSRT means nothing without a reference distribution. Each feature is converted to an age-banded z-score against the normative sample (8–10, 11–13, 14–16), then dimensions are formed as weighted means of their constituent z-scores.

$$z_i = (x_i - \mu_{\text{ageband}}) \div \sigma_{\text{ageband}} \quad | \quad D_k = \Sigma(w_i \times z_i) \div \Sigma w_i \quad | \quad \text{reported with 90\% CI from split-half reliability}$$

UNTIL NORMS EXIST

The normative parameters μ and σ do not exist before the validation study. Until then the pipeline outputs **raw features and sample-relative percentiles only**, and the L4 stage is disabled. Shipping z-scores against an imaginary norm is the most common failure of gamified assessment products.

7.2 What the researcher receives

OUTPUT	FORMAT	CONTENTS
Trial-level export	CSV / Parquet	One row per trial, all condition flags and QC columns — the file a statistician actually wants
Feature matrix	CSV	Participant × metric ID, with missingness codes and reliability estimates
Session QC report	PDF / HTML	Calibration values, attention-probe pass rate, exclusion flags, distress events
Codebook	Markdown / JSON	Every metric ID with its formula, units, direction, and known confounds
Reproducibility bundle	Archive	Scoring code version, schema version, stimulus set hash, analysis seed
Individual profile	Disabled in v1.0	Requires completed validation and ethics sign-off

7.3 A worked example of the inference chain

LAYER	CONTENT FOR ONE HYPOTHETICAL PARTICIPANT
L1 event	response · M1 · block=angry · is_nogo=true · rt_ms=291 · outcome=commission
L2 trial	Trial 37 valid (focus true, jitter 3.1 ms, RT above the 150 ms anticipation floor)
L3 feature	m1_commission angry = 0.38, neutral = 0.19 → m1_emo_cost = 0.19
L4 dimension	D2 Emotional inhibition cost = +1.4 SD versus the 11–13 band
Statement	“This participant withheld responses normally to neutral cues but failed to withhold substantially more often when angry faces were present, relative to same-age peers in this sample.”

**Not a
statement**

"This child is aggressive." The battery cannot support that sentence, and the software will not print it.

8 Psychometric validation plan

A gamified task is not a measure until it demonstrates reliability and validity in the population of interest. This section is the work that converts Skyhaven from a prototype into an instrument, and it should be pre-registered before the first participant is run.

8.1 Study design

PARAMETER	SPECIFICATION
Design	Cross-sectional validation with a two-week test–retest subsample (n ≈ 60)
Target sample	n ≈ 400 across three age bands, balanced by sex, mixed school types; powered to detect r = 0.20 at 80% power
Enriched subsample	n ≈ 80 with elevated teacher-rated aggression, for known-groups validity
Setting	Supervised, standardised devices where possible, quiet room, single session
Pre-registration	Hypotheses, metric definitions, exclusion rules and analysis plan registered before collection

8.2 Criterion measures collected alongside

MEASURE	INFORMANT	PURPOSE
Reactive–Proactive Aggression Questionnaire	Child self-report	Convergent validity, subtype separation — the key criterion
Strengths and Difficulties Questionnaire	Teacher and parent	Conduct problems criterion and the hyperactivity covariate for C7
Peer nomination or teacher rating of aggression	Teacher / peers	Known-groups validity, independent of self-report bias
Brief receptive vocabulary or reasoning task	Direct	Discriminant validity — scores must not merely track ability
Prosocial behaviour subscale	Teacher	Discriminant check on the retaliation metrics

8.3 Acceptance thresholds

PROPERTY	TEST	MINIMUM TO RETAIN A METRIC
Internal consistency	Split-half, Spearman–Brown corrected	$\rho \geq 0.70$ for RT metrics, ≥ 0.65 for choice indices
Test–retest	Two-week ICC	ICC ≥ 0.60
Convergent validity	Correlation with RPQ reactive subscale	$r \geq 0.20$, correct direction, $p < .01$
Subtype discrimination	Differential correlation with reactive vs. proactive subscales	Significant difference in dependent correlations
Discriminant validity	Partial correlation controlling ability and SDQ hyperactivity	Association survives adjustment
Fairness	Differential item/task functioning by sex, language, device class	No substantial DIF; otherwise the metric is revised or dropped

PRE-COMMITTED DECISION RULE

Any metric that fails its threshold is removed from the scoring pipeline and reported as a null result. This is stated in advance precisely so that a weak metric cannot be quietly retained because the product needs it.

8.4 Primary hypotheses to pre-register

ID	HYPOTHESIS	KEY TEST
H1	m6_hab correlates positively with reactive aggression, and more strongly with reactive than proactive.	Dependent-correlation comparison
H2	m1_emo_cost correlates with reactive aggression after controlling for neutral-block inhibition and SDQ hyperactivity.	Hierarchical regression
H3	m6_expectancy correlates preferentially with proactive aggression.	Dependent-correlation comparison
H4	Frustration deltas (m3_anger_delta, m3_ttq) add incremental variance beyond questionnaires.	ΔR^2 over criterion-only model
H5	Behavioural metrics show smaller social-desirability contamination than self-report, indexed by m6_sdr.	Moderation analysis

9 Ethics, consent, safeguarding and data protection

This battery deliberately induces mild frustration and provocation in minors and infers sensitive psychological characteristics. That combination places it firmly in the category of research requiring formal ethical review. Nothing in this section is optional.

9.1 Approvals required before any data collection

APPROVAL	SCOPE
Institutional ethics committee / IRB	Full protocol, stimuli, debrief script, distress procedure; M7 reviewed as a separate line item
School or institutional gatekeeper	Site permission, supervision arrangements, opt-out communication to all families
Data protection assessment	DPIA covering minors' behavioural data, retention, and cross-border transfer if any
Stimulus licensing	Documented research licence for the facial-affect set

9.2 Consent architecture

LAYER	REQUIREMENT
Guardian informed consent	Written, in the family's language, disclosing that some rounds are intentionally unwinnable and that mild frustration is part of the design
Child assent	Age-appropriate, spoken and on-screen, obtained at session start and again before M7
Ongoing assent	A visible "I want to stop" control on every screen; stopping is presented as costless and is never framed as failure
Withdrawal	Data deletable on request up to the point of anonymised aggregation, with the deadline stated in the consent form

9.3 Frustration exposure limits

CONTROL	LIMIT
Maximum consecutive rigged failures	2
Maximum time on any blocked trial	90 seconds, hard cap
Total frustration exposure per session	≤ 6 minutes cumulative
Exit control	Visible from the first trial, no penalty, no persuasion screen
Session ending	Always a winnable sequence plus mood-repair activity, regardless of route through the battery

9.4 Distress stop-rule and escalation

Triggered by any of: a self-reported anger or upset rating of 5 of 5; two consecutive ratings of 4; a supervising adult's flag; crying, verbal distress or attempted physical withdrawal. On trigger the system immediately halts the module, launches mood repair, delivers the debrief, records a `distress_flag` event, and notifies the responsible researcher. The child is not asked to continue. Where distress appears to reflect something beyond the task, the site's existing safeguarding referral pathway applies — the research team does not attempt to manage it informally.

9.5 Debrief script requirements

- State explicitly that some gates and setbacks were built so that **no one** could beat them, and that this was true for every child.
- Confirm the game was not testing whether the child is good, clever, or well behaved.
- Confirm that no score will be shared with teachers, classmates or parents as a judgement of the child.
- Offer a question opportunity and a genuinely winnable closing activity.
- Record completion as an event; a session without a completed debrief is a protocol deviation.

9.6 Data protection

CONTROL	IMPLEMENTATION
Pseudonymisation	Behavioural stores hold only participant_id; the identity vault is a separate system with separate access control
Data minimisation	No audio or video recording of the child unless the open-response step is separately consented; transcripts replace audio at the earliest point
Encryption	TLS in transit, encryption at rest, key rotation
Access control	Named researchers only; all access logged; no vendor or product-team access to identifiable data
Retention	Identifiable data destroyed at the stated deadline; anonymised research data retained per the approved plan
Prohibited uses	No advertising, no product profiling, no sharing with schools for discipline, no automated decision-making about any child

THE MISUSE RISK TO NAME EXPLICITLY

The most likely harm from this system is not distress during the session — it is a school receiving a list of “high aggression” children and acting on it. Behavioural indicators of this kind have modest effect sizes and substantial measurement error; used for individual labelling they would misclassify a large number of children. Governance, not accuracy, is the control that prevents this, which is why individual reporting is disabled in software rather than merely discouraged in a policy document.

10 Limitations and threats to validity

Stated plainly, because a reviewer will find them anyway and a protocol that names its own weaknesses is more credible than one that does not.

THREAT	DESCRIPTION	MITIGATION AND RESIDUAL RISK
Ecological validity	Tapping a rival’s lantern in a game is not hitting a classmate. Laboratory aggression analogues correlate with real aggression, but modestly.	Validate against teacher and peer report; residual risk remains and must be stated in any publication.
Construct non-specificity	Poor inhibition, low persistence and slow RT accompany ADHD, anxiety, fatigue, low motivation and hunger.	SDQ covariate, attention probes, standardised session timing; cannot be fully eliminated.
Gaming experience	Experienced gamers have faster RTs and greater tolerance for troll-style difficulty, independent of temperament.	Collect a brief gaming-frequency item and include it as a covariate.
Demand characteristics	Children infer that the game is “about anger” and adjust their behaviour.	Neutral in-world wording, cover story, and the m6_sdr index; partially mitigated only.
Cultural specificity of vignettes	What counts as provocation, and what response is appropriate, varies by culture and school context.	Local pilot and cognitive interviewing before use in any new population; norms are not transferable across contexts.
Single-session snapshot	Behaviour on one afternoon is not a trait.	Test–retest subsample quantifies stability; single sessions must not be read as dispositional.
Practice and fatigue	38 minutes is long for an 8-year-old; late modules may show fatigue effects.	Counterbalanced order, breaks, engagement index; consider a two-session split for the youngest band.
Self-report floor	Some children under-report anger even on pictorial scales.	Behavioural metrics are primary; affect ratings support interpretation rather than drive it.

10.1 Roadmap

PHASE	WORK	EXIT CRITERION
P0 Build	Implement M1–M6, event pipeline, calibration, QC	Deterministic replay of a session reproduces every feature exactly
P1 Usability pilot	n ≈ 20, think-aloud, comprehension checks	≥ 90% comprehension across age bands; no distress triggers attributable to unclear instructions
P2 Reliability pilot	n ≈ 60, split-half and retest	Metrics meeting §8.3 thresholds identified; failures dropped
P3 Validation	n ≈ 400 pre-registered study	Convergent and discriminant validity reported, norms constructed
P4 Controlled release	Enable L4 scoring; M7 only with approval	Published validation and an approved governance framework

Appendix A Metric formula reference

METRIC ID	DIM	FORMULA	UNITS / RANGE
m1_commission	D1	$n(\text{commission}) / n(\text{no-go})$	0–1, lower better
m1_dprime	D1	$z(H) - z(FA)$, log-linear correction for extreme rates	–3 to 5, higher better
m1_ssrt	D1	$RT_{[n]} - \text{mean}(SSD)$, $n = p(\text{respond} \text{signal}) \times N_{\text{goRT}}$	ms, lower better
m1_rtcv	D1	$SD(RT_{\text{go}}) / \text{mean}(RT_{\text{go}})$	ratio, lower better
m1_pes	D1	$\text{mean}(RT_{n+1 \text{error}}) - \text{mean}(RT_{n+1 \text{correct}})$	ms, near zero = poor monitoring
m1_emo_cost	D2	$m1_commission_{\text{angry}} - m1_commission_{\text{neutral}}$	–1 to 1, higher = anger-specific deficit
m2_lisas	D1	$\text{meanRT} + (SD_{RT}/SD_{PE}) \times PE$	ms-equivalent
m2_decay	D1	$\text{slope}(m2_lisas \sim \text{level})$	ms per level
m3_tttq	D3	$\text{median}(t_{\text{quit}} - t_{\text{trialstart}})$ over rigged trials	s, non-monotonic meaning
m3_effort_auc	D3	$\int \text{interactions } dt / (\text{cap} \times n_{\text{rigged}})$	0–1
m3_anger_delta	D3	$\text{anger}_{\text{post-rigged}} - \text{anger}_{\text{baseline}}$	–4 to 4
m4_persev	D3	mean consecutive old-rule responses after change point	trials
m5_severity	D5	$\Sigma w / (3 \times n_{\text{events}})$	0–1
m5_escalation	D5	$\text{slope}(w \sim \text{event order})$	weight per event
m5_latency	D5	median decision time where $w \geq 2$	ms
m6_hab	D4	$n(\text{hostile attribution}) / n(\text{ambiguous vignettes})$	0–1
m6_anger_amb	D4	mean anger rating, ambiguous vignettes	1–5
m6_expectancy	D6	$n(\text{aggressive choice rated "better outcome"}) / n(\text{aggressive choices})$	0–1
m6_sdr	cov	$ \text{severity}_{\text{open-ended}} - \text{severity}_{\text{forced-choice}} $	0–1
m7_repeat	D6	$n(\text{would repeat}) / n(\text{aggressive choices})$	0–1

Appendix B Data-quality and exclusion rules

Applied automatically at L2, before any feature is computed. Every exclusion is recorded with its reason code so that attrition can be reported.

LEVEL	RULE	REASON CODE
Trial	RT < 150 ms — anticipatory, not a decision	rt_anticipatory
Trial	RT > mean + 3 SD within participant and condition	rt_outlier
Trial	Frame jitter > 20 ms, or window unfocused during the trial or the one before	timing_invalid

Block	Accuracy at or below chance on go trials	block_chance
Block	p(inhibit) outside 0.25–0.75 on stop trials — staircase failed to converge, SSRT not computable	ssrt_invalid
Module	Practice mastery not reached in 3 attempts	not_comprehended
Module	M6 comprehension checks failed on > 25% of vignettes	sip_invalid
Module	Hostile attribution given in the clearly benign condition on all three items	indiscriminate_response
Session	> 20% attention probes failed	disengaged
Session	Calibration lag $\geq$ 150 ms — RT metrics dropped, choice metrics retained	rt_unreliable
Session	Distress stop-rule triggered	distress_stop
Session	Split-half reliability < 0.60 on primary metrics	unreliable_session

Appendix C Evidence base

The sources below are those underpinning the paradigms used. Citation details should be verified against the original records before publication; several were supplied as links in the source design and are reproduced here by their standard reference.

1. Berkowitz, L. (1989). Frustration–aggression hypothesis: Examination and reformulation. *Psychological Bulletin*, 106(1), 59–73. — basis for M3/M4; establishes that frustration produces negative affect with multiple possible outcomes, not automatic aggression.
2. Crick, N. R., & Dodge, K. A. (1994). A review and reformulation of social information-processing mechanisms in children’s social adjustment. *Psychological Bulletin*, 115(1), 74–101. — the six-step model structuring M6.
3. Dodge, K. A. (1980). Social cognition and children’s aggressive behavior. *Child Development*, 51(1), 162–170. — original ambiguous-provocation paradigm; the source of correction C1.
4. Orobio de Castro, B., Veerman, J. W., Koops, W., Bosch, J. D., & Monshouwer, H. J. (2002). Hostile attribution of intent and aggressive behavior: A meta-analysis. *Child Development*, 73(3), 916–934.
5. Verhoef, R. E. J., Alsem, S. C., Verhulp, E. E., & De Castro, B. O. (2019). Hostile intent attribution and aggressive behavior in children revisited: A meta-analysis. *Child Development*. — the 111-study, 29,272-participant synthesis cited in the source design.
6. Martinelli, A., Ackermann, K., Bernhard, A., Freitag, C. M., & Schwenck, C. (2018). Hostile attribution bias and aggression in children and adolescents: A systematic literature review. *Aggression and Violent Behavior*, 39, 25–32. — 27-study review linking HAB particularly to reactive aggression.
7. Arsenio, W. F., Adams, E., & Gold, J. (2009). Social information processing, moral reasoning, and emotion attributions: Relations with adolescents’ reactive and proactive aggression. *Child Development*, 80(6), 1739–1755. — basis for the outcome-expectancy metric distinguishing proactive aggression.
8. Study of emotional Go/No-Go performance in reactively aggressive adolescents (PMC7556161, 2020) — source of the angry-face inhibition contrast implemented as m1_emo_cost.
9. Logan, G. D., & Cowan, W. B. (1984). On the ability to inhibit thought and action: A theory of an act of control. *Psychological Review*, 91(3), 295–327. — horse-race model underlying SSRT.
10. Verbruggen, F., et al. (2019). A consensus guide to capturing the ability to inhibit actions and impulsive behaviors in the stop-signal task. *eLife*, 8, e46323. — the staircase and integration-method procedures in correction C3.
11. Dodge, K. A., & Coie, J. D. (1987). Social-information-processing factors in reactive and proactive aggression in children’s peer groups. *Journal of Personality and Social Psychology*, 53(6), 1146–1158.
12. Raine, A., et al. (2006). The Reactive–Proactive Aggression Questionnaire. *Aggressive Behavior*, 32(2), 159–171. — primary criterion measure in §8.2.

13. Goodman, R. (1997). The Strengths and Difficulties Questionnaire: A research note. *Journal of Child Psychology and Psychiatry*, 38(5), 581–586. — criterion and hyperactivity covariate.
14. Bradley, M. M., & Lang, P. J. (1994). Measuring emotion: The Self-Assessment Manikin and the semantic differential. *Journal of Behavior Therapy and Experimental Psychiatry*, 25(1), 49–59. — pictorial affect scales, correction C12.

CLOSING NOTE FOR THE REVIEWER

Every quantity in this document is defined as a function of stored events, every task is tied to a named paradigm, every known weakness is written down, and the outputs that could harm a child are switched off until validation justifies switching them on. Where the design departs from the literature it does so explicitly and says why. That is the standard this protocol asks to be judged against.